Next Bill Amount Prediction

Panel presentation | Use Case 2

Forecast the next bill.
Explain the estimate.
Prioritize unusual increases.

A customer-level forecasting and monitoring prototype

Historical evaluation: March-May 2025

Latest forecast: June 2025, outcomes unverified

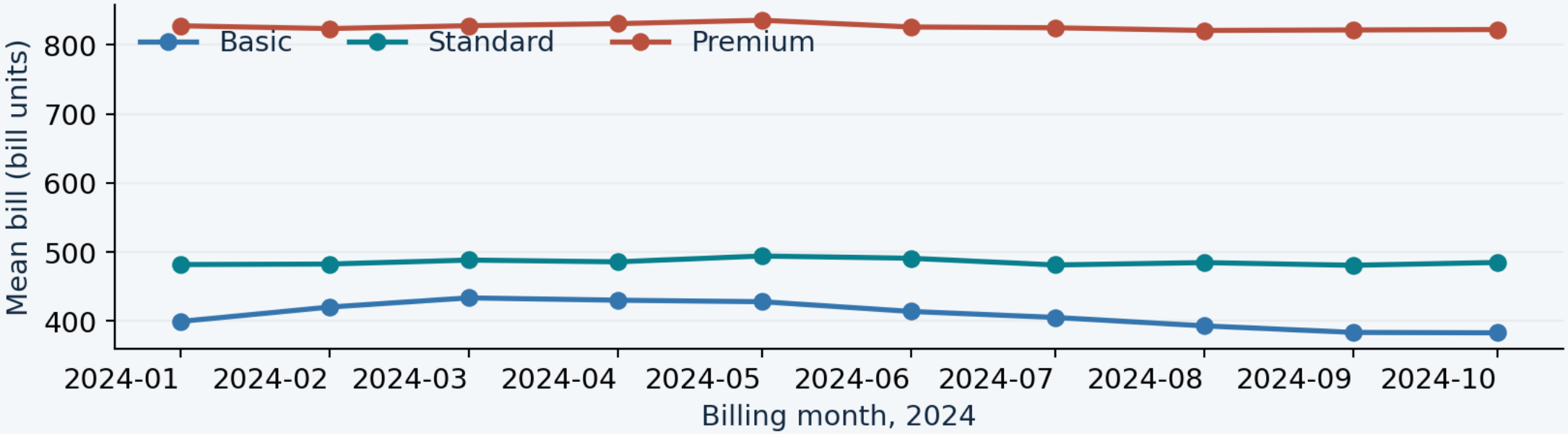
Business problem and decision

An early estimate helps teams investigate potential bill shock.

- Customers may face unexpected increases in recurring bills.
- Predict each existing customer’s next monthly bill after the current month closes.
- Give the operations team a review queue with historical comparisons, prediction ranges and influential factors.
- Support human review. Automatic bill changes and customer actions remain outside scope.

Data and historical billing patterns

17,000 records | 1,000 customers | 17 months | 15 raw columns



Training-period averages differ by plan. This short history cannot establish reliable annual seasonality. No missing values or duplicate customer-month keys were found.

Source: supplied raw CSV. Graph uses training-origin months Jan-Oct 2024. Amounts are in unspecified bill units.

Features use only information known at forecast time

Origin month M closes, then the model predicts month M+1.

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raw and engineered model inputs

3 months

recent bill and usage summaries

- Current information: plan, finalized usage, charges and current bill.
- History: prior bills, recent averages, volatility and month-to-month changes.
- Calendar: forecast-month sine and cosine features.
- Customer-specific rolling windows preserve chronology. Training data fits imputation, encoding and scaling.
- Exclude customer ID and next_bill_amount from the model inputs.

Source: notebook 01 and model_summary.json. Current-month fields require the documented close-of-month assumption.

Chronological evaluation matches monthly forecasting

The uploaded implementation uses the timeline below.

Purpose	Target months	Rows
Training	Feb-Nov 2024	10,000
Model selection	Dec 2024-Feb 2025	3,000
Point evaluation	Mar-May 2025	3,000
Latest forecast	Jun 2025	1,000

After selection, the regressor refits on 13,000 development rows. Preprocessing stays frozen from training. The fixed model uses newly observed histories at each monthly origin.

Monitoring later uses March for calibration and April-May for checks. Those months had prior point evaluation.

Gradient boosting gives the lowest validation MAE

Selection compares practical baselines with a small set of regression models.



Selected: 250 boosting iterations, learning rate 0.05, 15 leaves and L2 regularization 1. Random seed 42. No deep learning is required.

Sources: validation_model_comparison.csv, validation_baselines.csv and model_summary.json. Lower MAE is better.

The model improves MAE by 13.3%

Historical point evaluation: 3,000 rows, target months March-May 2025

50.96

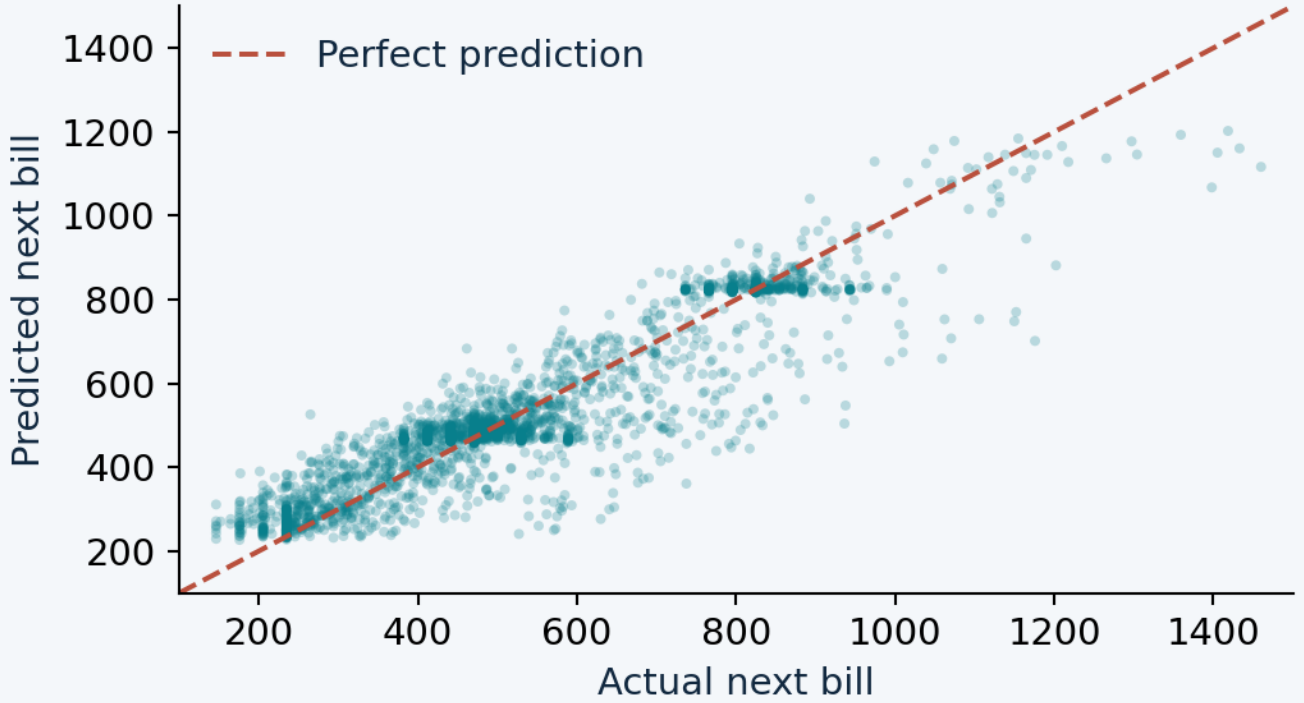
Model MAE

78.91

Model RMSE

9.54%

Model WAPE

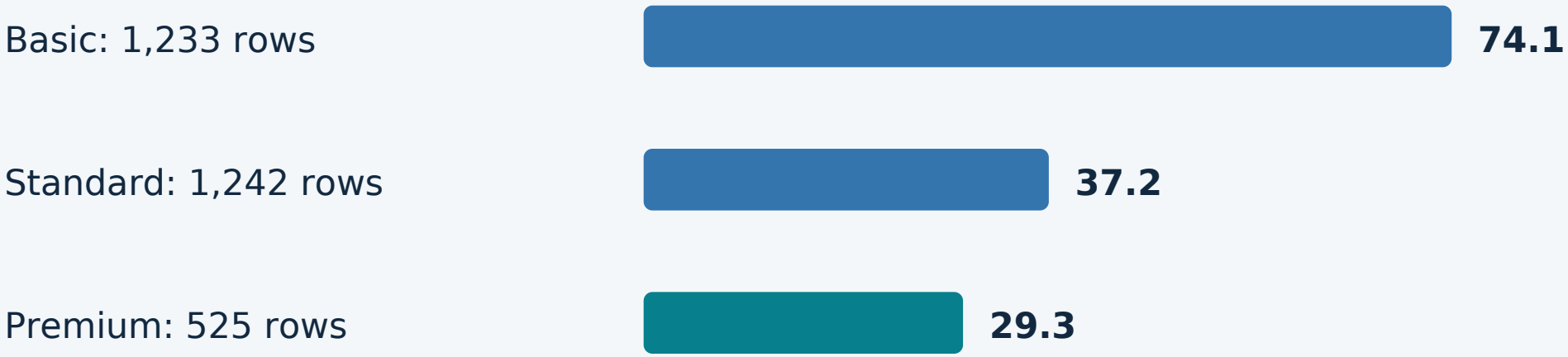


Recent-average baseline MAE: 58.80. Points on the dashed line are exact predictions. Extreme bills show larger errors and regression toward typical values.

Source: test_predictions.csv. Metrics independently recalculated from exported predictions.

Accuracy varies across plans

Basic-plan customers need the most attention in further error analysis.



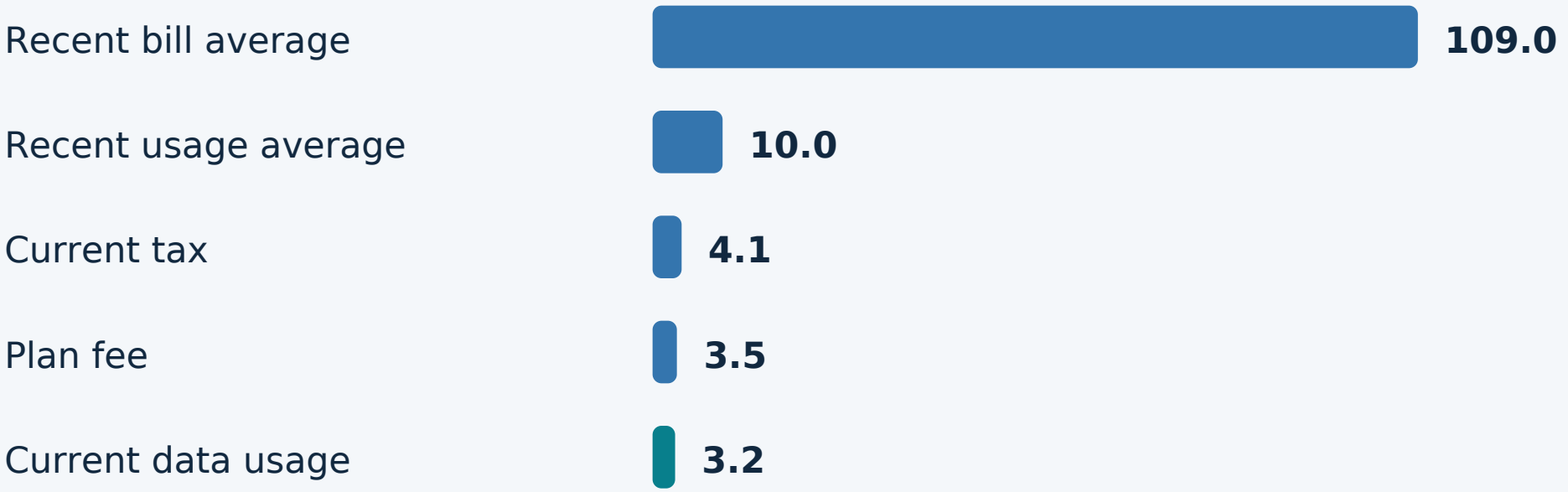
Plan WAPE: Basic 16.73%, Standard 7.42%, Premium 3.54%. Monthly MAE remains near 51 across March-May.

Lower Premium error does not imply strong within-plan fit: Premium R^2 is only 0.03. Review segment variation alongside absolute errors.

Source: test_segment_metrics.csv. Repeated customers contribute one row per evaluated month.

Recent billing history drives model predictions

Portfolio influence and customer explanations answer different questions.

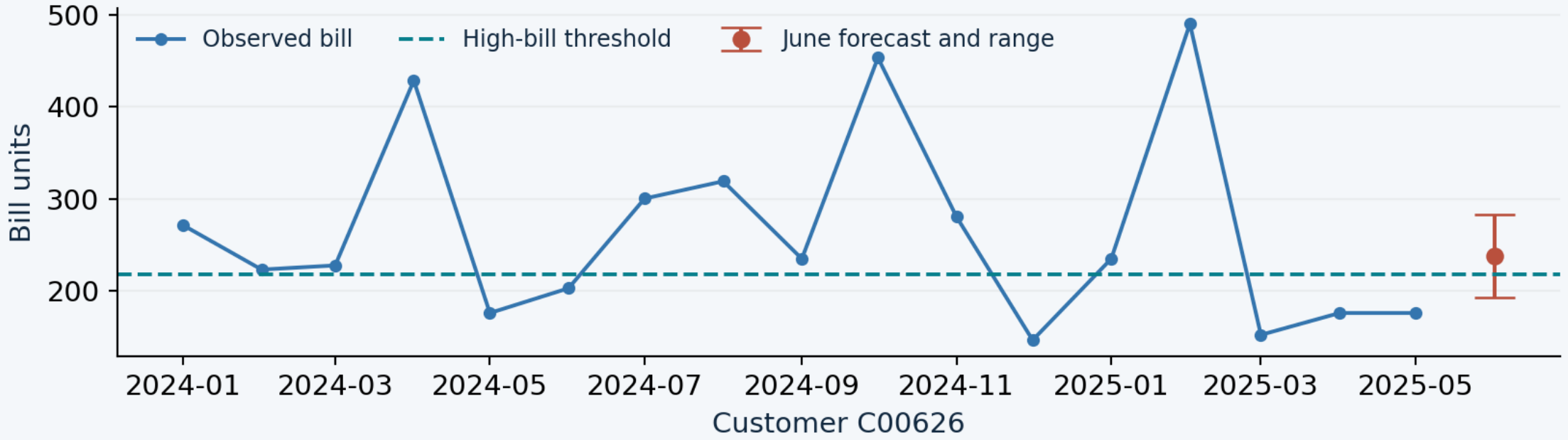


Bars show validation MAE increase after permutation. Customer explanations replace feature groups against a same-plan reference. Effects depend on group order and reference, and do not prove causality.

Sources: validation_feature_importance.csv and notebook 03. Correlated features can share or distort importance.

A customer forecast includes history and a range

C00626 | June prediction 237.79 | Range 192.37-283.21 | Threshold 217.90



March calibrates intervals. April-May empirical coverage is 88.35% against a 90% target. June outcomes are unverified. The point forecast crosses this customer’s high-bill threshold.

Sources: customer_history.csv and monitoring outputs. Prior evaluation exposure and dependence limit coverage guarantees.

High-bill monitoring shows a clear trade-off

Threshold = recent mean + max(20% of mean, 50 units, 2 × recent standard deviation)

Rule	Precision	Recall	Review load
Point alert	55.56%	2.87%	0.45%
Upper-range watchlist	10.51%	90.23%	74.70%

April-May contains 174 actual high bills. Point alerts catch 5. The wider watchlist catches 157 but flags 1,494 of 2,000 rows.

June review queue: 3 point alerts, 707 possible high bills and 290 within range. Sudden-spike detection remains a limitation. Thresholds are illustrative policy choices.

Sources: alert_quality.csv and monitoring_summary.json. Ground truth applies the same historical threshold to actual bills.

Prototype and final takeaway

A working project structure connects forecasting outputs to an operations view.

- Streamlit app: customer search, plan and priority filters, bill history, prediction range, drivers and CSV export.
- Demo path: open the review queue, select C00626, compare the forecast with history, then inspect model performance.
- The model improves average forecasting error. High-bill alerts still need better recall and a manageable review budget.
- Before deployment: validate fresh chronological outcomes, strengthen documentation and test changes in plans and usage.

Delivery: three analysis notebooks, model artifacts, exported predictions and app.py. App behavior was reviewed from source; live execution was not verified for this presentation.

Sources: app.py, notebooks 01-03 and supplied CSV exports in HCL-Mavericks-main.zip.